

PRAYAG NIKUL PURANI

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PROFESSIONAL EXPERIENCE

San Jose State University

Research and Teaching Assistance

San Jose, USA

Jan 2025 – May 2025

- Supported strategic research initiatives with advanced generative AI models, covered advanced techniques such as GANs, VAEs, Vision Transformers, Diffusion Models, fine-tuning LLMs, and Instruction Tuning maintaining data integrity and documentation of all records
- Designed a hierarchical multi-stage demographic classification system achieving 99% precision in ethnicity prediction from last names, leveraging character-level embeddings, LSTM, and NLP techniques; delivered a Flask web application with real-time and user interface
- Architected classification pipeline, starting with broad ethnicity detection using US Census surname data, then refining predictions for Asian subregions via USPTO datasets, enabling fine-grained demographic insights with enhanced model interpretability and scalability

C/test

San Jose, USA

ML Data Scientist

Jul 2024 – Dec 2024

- Translated cross-functional marketing goals an A/B testing assistant using GPT-4 to generate experiment variants (e.g., headlines, descriptions, CTAs) for email, ad, number of clicks, version comparison, and landing page tests, and accelerating campaign launches
- Merged Google Analytics 4 (GA4) user-level engagement data with Meta Ads campaign data to build a full-funnel attribution model
- Automated interpretation of test results using LLMs to summarize statistical outcomes, surface key takeaways, and suggest next actions significantly reducing data team overhead. Identified campaign inefficiencies that led to 10–15% improvements in conversion rates
- Reduced experimentation cycle time by 30% by eliminating manual bottlenecks in variant design, summary reporting, and stakeholder communication, created Tableau dashboards for product stakeholders, translating complex multi-touch attribution insights

Shree Hari Info Solution

Data Engineer

Ahmedabad, India

Jan 2022 – May 2023

- Objective: To find a potential solution using machine learning to help managers set coaching priorities to train their employees effectively by considering certain significant attributes that affect their performance
- Extracted, transformed and conducted Exploratory Data Analysis on the performance data of 7000+ representatives (20 million+ records) using SQL, Python (pandas, NumPy, statsmodels, scikit-learn, nitk, matplotlib, etc.) and R (MASS, ggplot)
- Created analytical hypothesis & validated it based on p-values(significance) by using predictive models to solve business problems, executed ETL pipelines from GCP, for employer performance, utilizing hypothesis testing and prediction model for decision making
- Optimized a real-time gesture recognition system using OpenCV, MediaPipe, and PyAutoGUI, achieving 90%+ accuracy in mouse control tasks for accessibility; leveraged computer vision pipelines for robust hand landmark detection and gesture classification

PROJECTS

PCB and Wafer Defect Detection.

Tech - AWS S3, AWS EC2, JIRA, Docker, Hugging face, Tensorflow, Pytorch, Flask, Jupyter Notebook, mAP, F1-Score, Precision, Recall, Accuracy

- Integrated 50GB+ multi-source industrial datasets into pipelines and managed progress via JIRA sprints, achieved 96.4% precision using YOLOv8, CNN, ViT and Faster R-CNN models, eliminated manual processing and detection
- Processed 10K image samples stored in AWS S3 using CRISP-DM, deployed on an EC2 instance with a Flask-based frontend, and built Tableau dashboards to visualize defect trends and KPIs, driving data-informed product quality decisions

Real-Time Smart Bank Data Streaming Capture.

Tech - Docker, AWS, Dynamic Data Generation, Real-Time pipeline, RestAPIs, Kafka, Pyspark, Elasticsearch, Flask, TimeStream

- Maximized a 99% capture rate by engineering a real-time transaction monitoring system using Kafka, PySpark, AWS IoT, and TimeStream leveraged Bloom filter to filter flagged transaction based on coordinates, and 3 optimized hyperparameters to enhance detection accuracy
- Implemented real-time dashboards with a 1s refresh rate using Grafana, Kibana, and Flask end-to-end interface; integrated CI/CD pipelines to enable seamless updates, and efficient data analysis for latitude, longitude and altitude, also flagged coordinates

SKILLS

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-Learn, Ultralytics, , PyTorch, TensorFlow, Keras), Java, Kafka, Spark, Hadoop, MySQL, MS Excel, Postgres, MongoDB, Snowflake, Redis, Cassandra, Neo4j, Athena, Redshift, Grafana, Kibana, Elasticsearch

Cloud Infrastructure: AWS, GCP, Azure, Microsoft Fabric, Docker, Cloud Computing, GitHub, Agile methodologies

Data Science: Predictive & Forecasting Modelling, Feature Engineering, ETL, Data Warehouse, Business and Product Analytics, Data Collection, Demand Forecasting, Operational KPI's, Anomaly Detection, Real-Time Processing, Advanced Analytics, Prompt Engineering

Machine & Deep Learning: Linear and Logistic Regression, Decision Tree, Random Forest, Transformers, CNN, ANN, NLP

LLM Tools & Frameworks: LangChain, OpenAI API, LLM Agents, LoRA, PEFT, Prompt Engineering, Evaluation, BERT, GPT, Ollama, PERT, LoRA, RAG, GenAI, HuggingFace Hub, Few-shot Learning, Zero-shot Learning, Chain-of-Thought (CoT), Self-Consistency

EDUCATION

San Jose State University

Master of Science in Data Analytics

Aug 2023 – May 2025

GPA 3.56/4

Relevant Courses: - Data Visualization, Big Data, Machine Learning, Deep Learning, Data Mining, Data Privacy, Generative AI models.

Vellore Institute of Technology

Bachelor of Engineering in Computer Science & Engineering

Jul 2019 – May 2023

GPA 8.69/10

Relevant Courses: - Artificial Intelligence, Data Structure, Statistics Engineers, Web mining, Cyber Security, Human Computer Interaction.